DMAIC Bootstrap: Improving an LLM-Agent Improvement System by Its Own Method

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Abstract

In March 2026 we showed that wiki-based AI agents can be created in minutes [1]. Operating those agents for several months told a different story: LLM-based agents systematically fail to follow a simple working-style triple — follow instructions, follow rules, use the knowledge graph. We applied the Six Sigma DMAIC cycle [2] to every agent incident, storing each record as a semantic subobject in a Semantic MediaWiki [3] knowledge graph [4]. By 2026-07-13 the corpus held 100 DMAIC records with recurrence chains of up to 17 occurrences of the same failure class within two weeks — evidence that the non-compliance is systematic, not incidental. The improvement system itself degenerated in the process: phase fields were abused as narrative blobs and the phase semantics drifted away from their Six Sigma meaning [5, 2]. An improvement system that needs improvement by its own method is a bootstrap: this paper reports the DMAIC Bootstrap experiment [6], in which the schema, the corpus, and the governance of the improvement system — including the policy under which this very text was authored — are reworked in auditable DMAIC rounds whose before/after statistics are the paper's raw data.

Keywords

DMAIC, LLM Agents, Semantic MediaWiki, Knowledge Graph

1. Introduction

The predecessor article [1] demonstrated that a family of wiki-grounded AI agents can be stood up "in no time": agent personas, their rules, and their working context are ordinary pages in a Semantic MediaWiki [3] family, so creating an agent is a documentation act. The months of operation that followed produced the finding this paper is about: the agents systematically deviate from the working-style triple that governs them —

1. follow instructions,
2. follow rules,
3. use the knowledge graph.

To make the deviations measurable rather than anecdotal, every incident is recorded as a DMAIC record: a semantic subobject with the five Six Sigma phases Define, Measure, Analyze, Improve, Control [2, 5] plus incident metadata (failure class, severity, recurrence count, first/latest date). The records live in the public context model AgentTeamContext [4]; the incident data itself is internal, its statistics are published.

Two things went wrong with this improvement system, and both are results rather than embarrassments. First, the corpus shows recurrence chains — the same failure class recurring 17, 11, and 8 times within two weeks despite documented rules, explicit callouts, and available graph data [6] — indicating that instructions, rules, and knowledge-graph grounding as such do not bind current LLM behavior. Second, the improvement system degenerated while being used: measure fields accumulated multi-incident narrative blobs, and the deployed schema documented Define as "what went wrong — the problem statement", although Define states how things *should* be, with criteria, as in the PDCA Plan phase [5]: objectives first, no problem at the start. An improvement system that needs improvement by its own method is a bootstrap — hence the experiment's name.

TODO: target venue

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The DMAIC Bootstrap experiment [6] therefore reworks the improvement system in rounds, where *each round is itself a DMAIC cycle* over the schema and the corpus, with SMW audit queries producing the before/after compliance numbers that a round's Measure and Control phases require. The experiment is self-applying in a second sense: the agents doing the rework are the agents being studied, so every incident during the experiment — including incidents in writing this paper — enters the corpus as data.

The contributions are:

1. a grounded DMAIC schema for LLM-agent incidents in Semantic MediaWiki, with per-phase evidence-link properties that work around SMW's page-level annotation granularity — unlike Wikidata [7], where every statement carries references, SMW attaches in-text link annotations to the page, never to the subobject (Section 2);
2. round-log statistics showing where instructions, rules, and knowledge-graph data fail to bind LLM behavior (Section 3);
3. an auditable governance pattern — issue-gated policy changes — under which the experiment's own rules, including the permission to author this text, are changed only through closable, citable gates (Section 4).

2. The DMAIC Bootstrap experiment

2.1. Corpus and baseline

At the start of the experiment (2026-07-13) the corpus held 100 DMAIC records across 18 data pages [6]. A census run that day found 155 page-level link annotations and 0 record-level citations: the rule "every claim cites its graph source" was not auditable at record level, because SMW in-text annotations attach to the page, not to the subobject. The largest recurrence chains were 17 (graph ignored in favor of ad-hoc probing), 11 (rules read but not applied when writing), and 8 (evidence available but not used); the human supervisor reported a callout rate of over 250 interventions in 14 days [6].

2.2. Schema: grounding as an auditable invariant

The schema round (R0) corrected the phase semantics in the context model [4] and added five per-phase evidence-link properties (dlinks, mlinks, alinks, ilinks, clinks; type Page) to the DMAIC subobject, so that grounding becomes a queryable invariant: the target is at least one evidence link per measure. A follow-up round (R0.1) resolved a taxonomy clash by human decision: an eight-class failure taxonomy grown from practice (A-overwrite, B-fabrication, C-evidence-ignored, D-environment, E-language, M-hidden-state, P-process, R-rule-substitution) moved to a dedicated `failureClass` property, while the previous category property became tag-like. The same round exposed and fixed a toolchain gap: multi-value separators were silently dropped between model and generated templates, requiring releases of the two upstream generator components before evidence links could round-trip from the declared model into queryable property values [6].

3. Rounds and results

Each round is a DMAIC cycle with a published round log [6]. Table 1 summarizes the rounds executed so far.

The exemplar round R1 also produced findings that shape the corpus rounds: an *attribution gap* (a seed record counts 6 cases while page-level grounding finds 8 candidate day pages in the range — exact attribution requires record-level review), *taxonomy pollution* (15 records carry ad-hoc one-off categories outside both taxonomies), and a representation limit (analysis evidence living on another wiki cannot be expressed in the Page-type link properties) [6].

Notably, the experiment recorded incidents in its own execution on its first day: the rounds R0/R0.1 were executed before the paper repository existed — an inversion of the house Document First rule

Table 1
DMAIC Bootstrap rounds as of 2026-07-13 (source: round logs [6]).

round	date	result
R0	2026-07-13	phase semantics corrected; five evidence-link properties added; baseline measured: 100 records, 0 grounded on any phase
R0.1	2026-07-13	taxonomy decision (failureClass vs. tags); two upstream toolchain fixes released; separators round-trip model→generator
R0.2	2026-07-13	Document First restored (paper repository created); publisher-independence enforced after human callout
R1	2026-07-13	seed triple grounded: per-phase evidence links set from graph queries; grounding audit 0 → 3/3/0/3/3 (define/measure/analyze/improve/control) of 103 records

— and an agent attempted to use a platform-default persistence mechanism in place of the mandated wiki-based one. Both entered the corpus as R-rule-substitution records [6]: the instruments observing the failure modes exhibit the failure modes.

4. Governance as experiment data

Because the experiment studies agent-authored artifacts, the rules under which agents author are part of the observed system. The paper repository therefore started with a strict no-scientific-content policy for AI, and every policy change is issue-gated: a dedicated issue must be closed by the human authors, with the policy amendment in the same commit that closes the issue. Two gates have been exercised so far: the publisher-independence gate (no venue claims before an outlet decision; the working layout is venue-neutral in substance) and the scientific-content gate, opened on 2026-07-13, under which this text is authored — grounded in the round logs and public experiment pages, with human review and human responsibility for all claims. The gates, their timing, and their resolutions are citable experiment data in the repository's issue tracker.

Declaration on Generative AI

The authors used an LLM-based agent (Claude, model claude-fable-5) to author parts of this paper — abstract, introduction, experiment description, and this declaration — under the issue-gated content policy of the paper repository: agent-authored content must be grounded in the experiment's round logs and public pages with every claim citing its source, and enters only via commits reviewed by the human authors. The agent's behavior while producing the paper is itself part of the reported experiment data. The human authors reviewed and edited the content and take full responsibility for the publication's content.

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